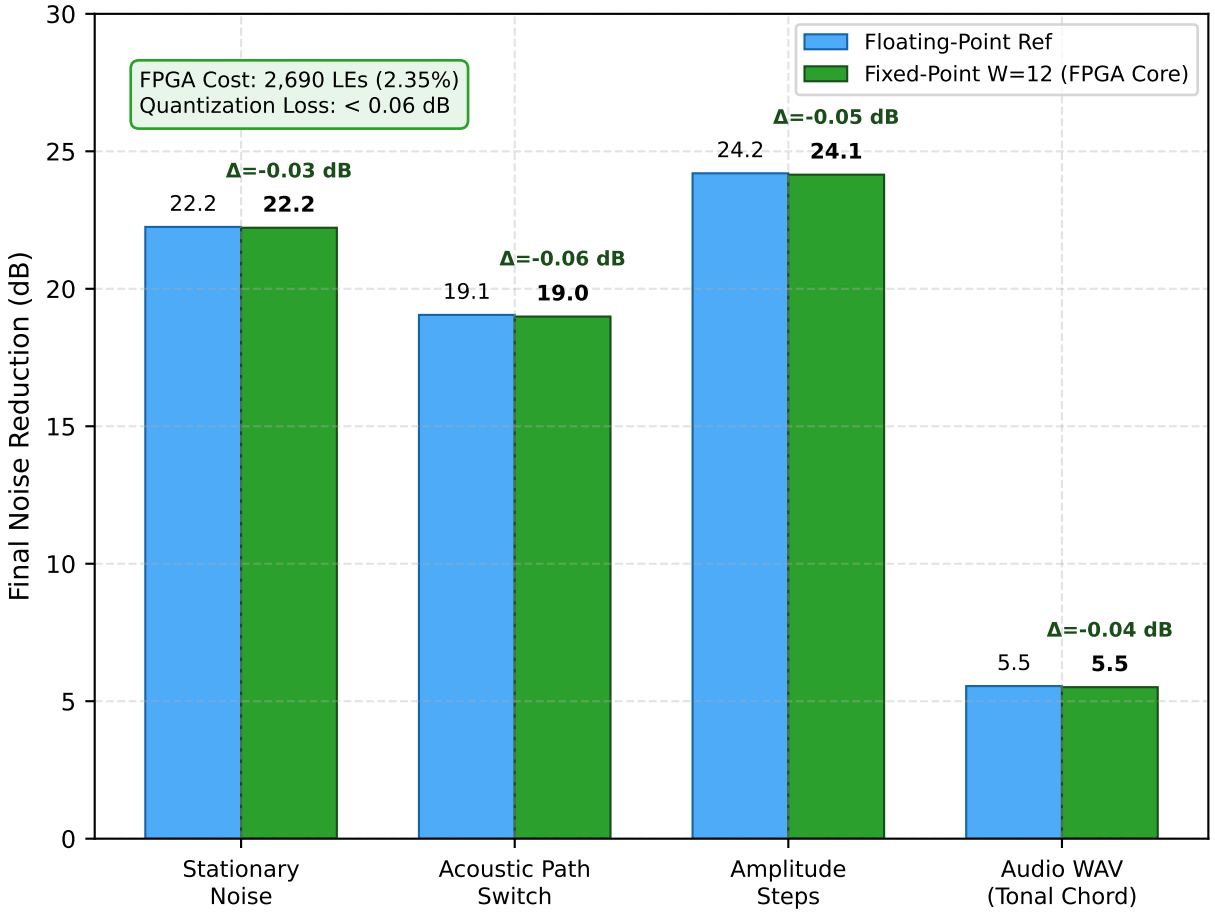

FPGA Implementation Status & Complexity Comparison (W=12)
Cyclone IV E EP4CE115F29C7 (Speed Grade 7)

NLMS-Shift (W=12, Dividerless)
SYNTHESIZED (Quartus Prime 25.1)
• LEs: 2,690 (2.35%) • Regs: 1,213 • Mults: 4
• Fmax: 61.98 MHz • Setup Slack: +3.867 ns

Standard NLMS (W=12, Restoring Divider)
SYNTHESIZED (Quartus Prime 25.1)
• LEs: 2,697 (2.36%) • Regs: 1,425 • Mults: 4
• Fmax: 61.72 MHz • Setup Slack: +3.798 ns

LMS (W=12, Fixed Step Size)
SYNTHESIZED (Quartus Prime 25.1)
• LEs: 1,885 (1.65%) • Regs: 1,148 • Mults: 4
• Fmax: 59.90 MHz • Setup Slack: +3.305 ns

Hardware Pareto Summary (W=12): All three architectures achieve full timing closure (>50 MHz) with positive setup slack >= +3.3 ns on Speed Grade 7 silicon. NLMS-Shift achieves the highest Fmax (61.98 MHz) and lowest latency (72 cycles) while eliminating divider stalls.